

3D Scanner

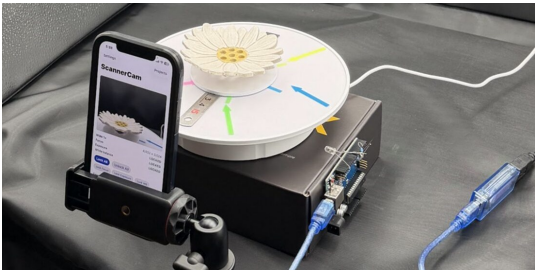
Arie Meir designed and built an end-to-end tabletop photogrammetry rig using an iPhone 17 Pro Max, Arduino, Mac and commodity hardware, with **Claude Code** used extensively as an **AI engineering agent**. The goal: turn a real ceramic form into editable, printable geometry and test how far AI could accelerate cross-domain hardware/software integration.

The problem

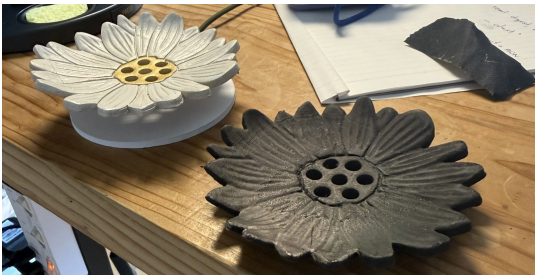
An artist wanted to make larger versions of a commercially molded clay flower and explore deeper forms without changing the diameter. The original geometry existed only as a physical object. A useful path forward required capturing that surface accurately enough to modify, re-fabricate and eventually use as the basis for new tooling.

The approach

Instead of buying a dedicated scanner, the rig coordinates a motorized turntable, an Arduino that impersonates its IR remote, and a custom iPhone camera server. A Mac drives the scan loop, **Apple Object Capture** reconstructs the mesh, and PyMeshLab / Blender turn the result into a printable STL.



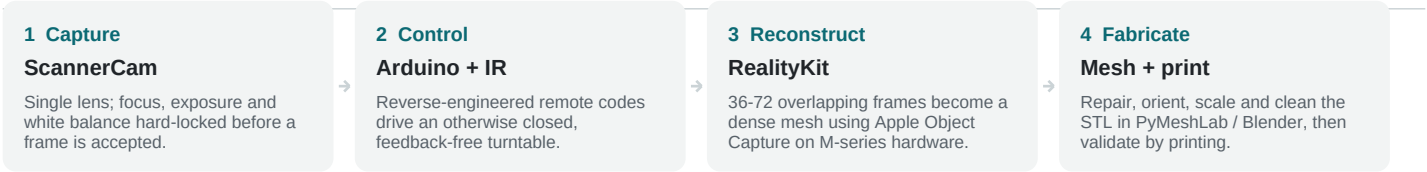
The capture rig: iPhone camera server, turntable and Arduino IR control.



Original ceramic flower and reconstructed FDM reproduction.

Impact

The prototype proved the complete **physical-to-digital-to-physical** loop: the ceramic flower was scanned, reconstructed, cleaned, dimensioned and reproduced as the black 3D print. That creates a modifiable digital starting point for larger or deeper mold variants rather than being locked to the commercial tool.



Engineering architecture

Unobservable actuator state

The turntable exposes only a START/PAUSE toggle and no position feedback. Any ambiguous command drops the controller to **UNKNOWN** and halts the scan rather than guessing.

Verified capture chain

Each JPEG is **SHA-256** verified during transfer. Idempotent request IDs prevent duplicate shutters, and partial downloads are never promoted to valid frames.

Photogrammetry-safe camera

The app pins one physical lens and can reject capture unless focus, exposure and white balance are locked, preventing silent optical changes between frames.

AI as engineering harness

Claude Code was used across Swift, Python, Arduino, photogrammetry and mesh tooling to accelerate integration and debugging; physical tests and system-level decisions closed the loop.

Technical scope

iOS / Swift camera server · HTTP API · Arduino IR firmware · Python scan orchestration · Apple RealityKit Object Capture · PyMeshLab + Blender · FDM print validation · AI-assisted cross-domain integration

Design principle: use AI to cross domain boundaries quickly, but make physical uncertainty explicit and validate every transition from actuator to image to mesh to manufactured part.